

awk Command – Practical Linux Notes

What is awk?

- awk is a **text processing tool + mini programming language**
- Mostly used to:
 - Extract columns (fields)
 - Filter lines
 - Modify text
- Commonly used with pipes (|) on command output

How awk Works (in short)

- Reads input **line by line**
- Splits each line into **fields**
- Default field separator: **space**
- Fields are accessed using:
 - \$1, \$2, \$3 ...
 - \$0 → entire line

Check awk Version

awk --version

✓ Shows installed awk version

Print Specific Fields from a File

awk '{print \$1}' file

✓ Prints **1st column** from each line

awk '{print \$1, \$3}' file

✓ Prints **1st and 3rd column**

Using awk with ls -l

ls -l | awk '{print \$1, \$3}'

✓ Prints:

- \$1 → file permissions
- \$3 → file owner

ls -l | awk '{print \$NF}'

✓ Prints **last field** (usually filename)

Search for a Word (Pattern Matching)

awk '/Jerry/ {print}' file

✓ Prints lines containing **Jerry**

Shortcut (same result):

awk '/Jerry/' file

Change Field Separator (-F)

awk -F: '{print \$1}' /etc/passwd

✓ Uses : as delimiter

✓ Prints **usernames** from /etc/passwd

Modify Fields (Replace Words)

```
echo "Hello Tom" | awk '{$2="Adam"; print $0}'
```

✓ Replaces **Tom** → **Adam**

```
cat file | awk '{$2="Imran"; print $0}'
```

✓ Replaces **2nd column** with Imran in every line

Filter by Line Length

```
awk 'length($0) > 15' file
```

✓ Prints lines longer than **15 characters**

Conditional Statements

```
ls -l | awk '{if ($9 == "seinfeld") print $0}'
```

✓ Prints only the line where:

- filename (\$9) is **seinfeld**

Count Number of Fields

```
ls -l | awk '{print NF}'
```

✓ Prints number of columns in each line

Important awk Variables (Practical)

Variable	Meaning
\$0	Entire line
\$1..\$n	Field numbers
NF	Number of fields
NR	Line number
FS	Field separator
OFS	Output field separator